

Group 2 — Superposition geometry

Where this sits. This is a per-group deep dive promised by the *Landscape* survey (its Group 2 paragraph and Part III/IV). It expands the five “forward-geometry” papers into one standalone treatment. Conventions are inherited from the *ML Concepts* primer: *feature* = direction; *superposition* present iff the embedding map is non-injective; *interference* = off-diagonal of the Gram matrix D^*D (adjoint $\Rightarrow$ metric-dependent); the residual stream carries a $GL(V)$ gauge with no privileged basis. Notation: W^* adjoint (metric invoked), $W^\vee$ dual (metric-free). All five papers are from the Anthropic / Transformer Circuits lineage; Part V of the *Landscape* survey flags the resulting provenance bias, which applies in force here.

The shared question

Group 3 (SAEs) and Group 4 (attribution) both ask an *inverse* question: given a trained model, recover the features and the circuit. Group 2 asks the *forward* question, and it is the only group in the corpus that can answer cleanly, because it works in a regime where the ground truth is known by construction. Fix a set of n idealized features, each with a sparsity S_i (probability of being inactive) and an importance I_i (a loss weight). Hand a network only $d \ll n$ dimensions to store them in. *What packing does gradient descent choose, and what does that packing cost?*

The thesis binding these five papers is that this is a well-posed geometric optimization problem with surprisingly rigid answers. Sparsity is the resource that makes packing more than d features worthwhile: a feature that is almost always zero almost never collides with another, so the *interference* it creates is rarely paid. Importance is the resource that decides *which* features get packed and which get dropped. The trade-off between them is not smooth — it is a *phase transition* — and the geometry on the favourable side of the transition is not arbitrary: features arrange into the vertices of uniform polytopes, the solutions of a generalized Thomson problem. The capacity of the channel is governed by the same mathematics — Johnson–Lindenstrauss and compressed sensing — that the *Landscape* survey identifies as the field’s shared substrate (Part I.2).

The five papers are five cross-sections of this one object. Elhage et al. [2] is the foundational study: the toy ReLU autoencoder, the phase diagram, the polytope geometry, the capacity bound. Henighan et al. [5] changes one knob — finite rather than infinite data — and finds that the same machinery now packs memorized *data points* instead of features, producing double descent. Olah et al. [6] steps back to ask what “distributed representation” even means, and splits it into two competing ways to spend the same exponential representational volume. Elhage et al. [4] attacks the load-bearing assumption underneath all of it — that the storage space has no privileged basis — and finds the symmetry is empirically broken, fingering the optimizer. Anthropic Interpretability Team [10] carries the geometry one level up, from features to the *weights between* features, and shows those too are forced into superposition, manufacturing spurious “interference weights”. They belong together because they share a single mathematical object — the low-rank Gram matrix W^*W and its off-diagonal — and disagree only about which cross-section of it to study.

The papers

Toy Models of Superposition

Elhage et al. [2] is the keystone. The setup is deliberately minimal. Synthetic “features” are a sparse non-negative vector $x \in \mathbb{R}^n$: each x_i is 0 with probability S_i , else uniform on $[0, 1]$. An embedding map $W : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (with $m \ll n$) compresses to a hidden vector $h = Wx$, and the model attempts to reconstruct through the *transpose* of the same map plus a ReLU:

$$h = Wx, \quad x' = \text{ReLU}(W^*Wx + b).$$

The loss is importance-weighted mean squared error, $L = \mathbb{E}_x \sum_i I_i (x_i - x'_i)^2$. The columns $W_i = We_i$ are the feature directions; whether feature i is represented at all is read off $\|W_i\|$, and whether it shares its space with others is read off $\sum_{j \neq i} (\hat{W}_i \cdot W_j)^2$.

The crisp definition the rest of the corpus inherits: a linear representation **exhibits superposition iff W^*W is not invertible** (equivalently W non-injective, the columns linearly dependent) — forced the moment $n > m$. The companion observation, which makes the geometry tractable, is the **correspondence between polytopes and low-rank PSD matrices**: any matrix of the form W^*W with W an $m \times n$ matrix is the Gram matrix of n points in $\mathbb{R}^m$, so every superposition strategy *is* a configuration of n points and vice versa.

Two competing forces govern the optimum. Decomposing the loss by sparsity pattern via the binomial expansion of $((1 - S) + S)^n$, the dominant term as $S \rightarrow 1$ is the 1-sparse loss L_1 . Under uniform importance, unit-norm features, and $b = 0$, L_1 splits into a **feature-benefit** term (rewarding representing a feature at all) and an **interference** term that is exactly a **generalized Thomson problem**: packing points on the sphere S^{m-1} to minimize a pairwise-interaction energy. The ReLU makes *negative* interference free in the 1-sparse case, which is why solutions prefer antipodal arrangements and a slight negative bias (it converts small positive interference into harmless negative interference). For the linear baseline $x' = W^*Wx + b$ the interference term is $\sum_{i \neq j} |W_i \cdot W_j|^2$ (the ℓ^2 analogue of compressed-sensing *coherence* $\max_{i \neq j} |W_i \cdot W_j|$, the ℓ^∞ version), and it is never worth representing more features than dimensions — the linear model does PCA. The single ReLU changes the answer qualitatively.

There is a genuine **phase transition**. For two features in one dimension, the natural strategies are $W = [1, 0]$ (drop the extra feature), $W = [0, 1]$ (dedicate the dimension), and the antipodal $W = [1, -1]$ (superpose, sacrificing the ability to represent both at once). The closed-form losses cross, giving a *first-order* phase change in the sparsity–importance plane: the optimal configuration jumps discontinuously. Three features in two dimensions adds richer choices, all describable as “which direction does W refuse to represent?” — e.g. $W \perp (1, 1, 1)$ yields the equilateral triangle, and more generally $W \perp (1, 1, \dots, 1)$ yields the regular simplex, the “minimal superposition” that declines only the fully dense direction.

The geometry deepens under **uniform superposition** (all features equal). The order parameter is dimensions-per-feature $D^* = m/\|W\|_F^2$, which is “sticky” at 1 and $1/2$. Resolving it per-feature via the **feature dimensionality**

$$D_i = \frac{\|W_i\|^2}{\sum_j (\hat{W}_i \cdot W_j)^2}$$

reveals that features cluster at specific fractions, each a named polytope: $1/2$ antipodal pair, $2/3$ triangle, $3/4$ tetrahedron, $2/5$ pentagon, $3/8$ square antiprism — precisely the small Thomson-problem

solutions. Many decompose as **tegum products**: orthogonal direct sums of lower-dimensional uniform polytopes, with *zero* interference across tegum factors (this is why the feature-interaction graph splits into disconnected components). So superposition is not one thing but a discrete ladder of geometric phases — “many phases of ice.”

The paper also shows the model can *compute* in superposition, not merely store: an untied model $h = \text{ReLU}(W_1x)$, $y' = \text{ReLU}(W_2h + b)$ learns $\text{abs}(x_i) = \text{ReLU}(x_i) + \text{ReLU}(-x_i)$ for more features than it has neurons, via an “asymmetric superposition” motif that turns dangerous positive interference into tolerable negative interference. On capacity, the **compressed-sensing** bound: an n -dimensional k -sparse vector is recoverable from $m = \Omega(k \log(n/k))$ measurements, read here as an *upper bound* on superposition. With density fixed by S (so $k = O((1 - S)n)$), this becomes $m = \Omega(-n(1 - S) \log(1 - S))$: the number of features is *linear* in m but modulated by sparsity, with the information-theoretic reading that $\log(1 - S)$ is the surprisal of a coordinate being nonzero. Superposition is also tied to **adversarial vulnerability**: the off-diagonal ϵ entries of $(W^*W)_i = (1, \epsilon, -\epsilon, \dots)$ are exactly an attack surface, and vulnerability scales with features-per-dimension.

Superposition, Memorization, and Double Descent

Henighan et al. [5] keeps the ReLU-output model but trains on a *finite* dataset of T examples, $X \in \mathbb{R}^{n \times T}$ (inputs normalized to $\|x\| = 1$). The reframe: visualized through feature columns W_i , finite-data solutions look like noise; visualized through the *hidden datapoint vectors* $h_i = WX_i$, they form clean **regular T -gons**. The model packs *memorized data points* in superposition, treating each training example as a one-off “datapoint feature.” Extending feature dimensionality to data points,

$$D_{X_i} = \frac{\|h_i\|^2}{\sum_j (\hat{h}_i \cdot h_j)^2},$$

gives $D_{X_i} = m/T$ for a T -gon, exactly as observed in the small-data regime. As T grows the model switches from storing T data points to storing the ~ 5 generalizing features (the pentagon of the infinite-data model). **Double descent** appears at the crossover — test loss rises before falling — and notably *without label noise*: the “noise” is the unavoidable reconstruction error of compressing $n = 10,000$ features into $m = 2$. A commenter’s analysis locates the generalization onset at $T(1 - S) \sim 10$ (each feature seen $\sim$ ten times), and a repeated datapoint is memorized once its repeat count $R \gtrsim T/N$. The deep point: “feature” and “memorized datapoint” are the *same* representational primitive viewed at different dataset sizes, and overfitting is mechanistically datapoint-superposition.

Distributed Representations: Composition & Superposition

Olah et al. [6] is conceptual, using Thorpe’s binary shape/colour codes to separate two ideas conflated under “distributed representation.” **Composition** represents an object as a combination of independent features (a “green” direction plus a “square” direction); n features compose into $\sim \exp(n)$ combinations, giving non-local generalization (data about green circles helps green squares) and interpretability (a linear number of parts). **Superposition** packs $\sim \exp(m)$ mutually-exclusive non-composing items into m neurons. The clean trade-off: m bits afford $\exp(m)$ “volume”; n *arbitrarily composable* features need $\exp(n)$ volume, so only m fully composable features fit, whereas with no composition one stores $\exp(m)$ items in superposition (a nonlinearity required to read

them out). **Composition and superposition are two ways to spend the same exponential volume, and they are in tension.** They reconcile only through sparsity: limited composition leaves low-probability “holes” in the compositional space that superposition can fill — an informal restatement of compressed sensing. This frames Toy Models’ sparsity axis as secretly a composition axis.

Privileged Bases in the Transformer Residual Stream

Elhage et al. [4] interrogates the assumption underpinning the whole group. The residual stream is read and written only by full-rank linear maps, so conjugating by any $M \in GL(V)$ (read maps $\mapsto W_{\text{in}}M^{-1}$, write maps $\mapsto MW_{\text{out}}$) gives a bit-identical function: **no privileged basis**, hence individual coordinates should be meaningless and a feature should smear $\sim 1/\sqrt{d}$ of its magnitude across each coordinate. Empirically false: large transformers show *outlier* coordinates up to $20\times$ typical magnitude. The diagnostic is the **kurtosis-3 test**. If features were random directions, an activation coordinate would be a sum of (rescaled) isotropic-Gaussian components, hence Gaussian, with **kurtosis** 3; the observed residual stream has kurtosis $\gg 3$ (heavy tails = basis privilege). A fixed random rotation restores kurtosis to ≈ 3 , confirming the metric is the diagnostic. The paper rules out LayerNorm (RMSNorm-style basis-free normalization does not fix it) and floating-point precision (the forward pass is genuinely rotation-invariant to within tiny noise; training in independent random bases per layer barely changes loss). The remaining culprit is **Adam**: its *per-coordinate* second-moment normalization is the one strongly basis-dependent operation left, and it apparently induces the layers to “collude” on a shared computation basis. This is the empirical fact that licenses the rest of the group to treat the standard inner product as meaningful — a crutch, not a principle.

A Toy Model of Interference Weights

Anthropic Interpretability Team [10] lifts the geometry from features to the weights *between* features. Reinterpret the toy model $x' = \text{ReLU}(W^*Wx + b)$ as an identity circuit put in superposition; the observed “downstairs” **virtual weights** are $U = W^*W$, the effective feature-to-feature couplings. If two layers hold features in superposition, the weights between them are *forced* into superposition too, manufacturing **interference weights**: spurious couplings that are pure noise — they do not help the loss and the model does not “want” them, yet they genuinely exist (and may be a source of adversarial examples). Two principled definitions: (1) **ideal weights** U^* are what you would get by finetuning the lifted feature-feature matrix with a small regularizer, and the interference weights are $U - U^*$; (2) the interference weights are those U_{ij} whose ablation barely changes the loss, $\Delta L(U_{ij}) < \epsilon$. Both reproduce the phenomenology of Bricken et al. [3]’s logit weights (a histogram of real-plus-noise weights; a scatter of two seeds where real weights agree and interference weights do not), with the hard case being real and interference weights that *overlap in magnitude* (so thresholding fails). Both definitions are intractable at scale (one needs $\sim n^2$ matrices or $\sim n^2$ ablations), motivating cheap heuristics (raw magnitude, expected/target-weighted attribution, frequency). A tempting compressed-sensing approach to “de-superpose” the weights is blocked because the lift need not be linear. The central distinction this paper draws: **real weights vs. interference weights** — separating signal from the noise that superposition forces into the circuit. The authors suspect whether interference weights can be ignored is the pivotal question for whether *global* (rather than per-prompt) circuit analysis is possible at all.

Common mathematical core

Strip the five papers down and one object remains: the low-rank Gram matrix $G = W^*W$ (equivalently D^*D for a dictionary D with unit columns), an endomorphism of feature-coordinate space whose diagonal is feature norms and whose off-diagonal is **interference**. Every quantitative claim is a statement about G .

- **Existence** of superposition is the metric-free statement $\ker W \neq 0$ (G singular), forced by $n > m$. This is linear dependence, nothing more.
- **Cost** is metric-dependent: it lives in the *off-diagonal* of G , which is built from the adjoint W^* and so presupposes an inner product. Toy Models’ interference $\sum_{j \neq i} (\hat{W}_i \cdot W_j)^2$, the coherence $\max_{i \neq j} |W_i \cdot W_j|$, the adversarial ϵ entries of $(W^*W)_i$, and the interference weights $U - U^*$ are all the same off-diagonal seen at different resolutions.
- **Near-orthogonality** ($G \approx I$) is what *sparsity* buys: features rarely co-activate, so off-diagonal entries are rarely paid. Its feasibility is Johnson–Lindenstrauss (exponentially many near-orthogonal directions exist in $\mathbb{R}^m$); its capacity limit is compressed sensing ($m = \Omega(k \log(n/k))$); its discrete optima are Thomson configurations (uniform polytopes, tegum products), since minimizing the interference energy on the sphere *is* the Thomson problem.

This is the connective tissue of the *Landscape* survey’s Part I (superposition as overcomplete almost-orthogonal frame; existence metric-free, cost metric-laden) and Part III (almost every quantity is a bilinear pairing; the metric-dependent ones secretly insert $G = I$). Group 2 is the place that substrate is studied head-on and in closed form.

How they diverge

The five agree on the object and split on which cross-section, and the splits are not cosmetic.

Infinite-data feature geometry vs. finite-data memorization. Toy Models and the composition note study the $T = \infty$ limit, where G ’s columns are *generalizing* features arranged as polytopes. The double-descent paper studies finite T , where the same geometry instead packs *datapoints* (T -gons in h -space). The order parameter shifts from feature dimensionality D_{f_i} to datapoint dimensionality D_{X_i} , and the governing variable from sparsity S to dataset size T (with the transition at $T(1 - S) \sim 10$). Same machine, different inputs, qualitatively different “features.”

Representation-level vs. weight-level superposition. Four papers study superposition of *activations*: features packed into one space, the object being G as a feature Gram matrix. The interference-weights paper studies superposition of *weights*: the couplings $U = W^*W$ between two superposed layers, the object being a feature-to-feature *map*. This mirrors the *Landscape* survey’s representation-vs-computation fork (Part IV.2): a Gram matrix describes a single space; a virtual-weight matrix describes a map between spaces, and the noise it inherits (interference weights) is a different, downstream pathology from the interference in a single layer’s frame.

The basis-privilege diagnostic is a different question entirely. Four papers *assume* a working inner product (the polytope angles, the T -gon radii, the ΔL ablations all presuppose one) and study geometry within it. Elhage et al. [4] asks whether that inner product is even legitimate — whether training breaks the $GL(V)$ gauge — and answers empirically yes, blaming Adam. It is not a packing result; it is the meta-result that licenses the others to speak of “near-orthogonal” at all. Without it, every angle in the group is gauge-arbitrary.

What they answer, and what they don’t

What is genuinely settled. Superposition is a real, demonstrated phenomenon in trained ReLU networks, not merely a useful post-hoc story: in these toy models it is the ground truth. The existence criterion (W^*W singular) is exact. The sparsity–importance phase transition is real and, in the smallest cases, has closed-form crossing loss curves confirming it is first-order. The capacity bound is a rigorous specialization of compressed sensing. Memorization-as-datapoint-superposition is a clean mechanistic account of double descent. The kurtosis-3 test is a sound, threshold-free diagnostic with a clear null.

What is honestly open. Closed forms exist *only* in toy regimes (the $n = 2, m = 1$ case and a handful of small uniform-polytope configurations); the authors explicitly list “are there models of superposition with closed-form solutions?” as open. Whether the toy geometry *transfers* to real models is unresolved and the authors hedge hard: superposition and polysemanticity probably generalize, but “the geometry and learning-dynamics results are much more uncertain” and “too elegant to be true.” There is **no statistical test for superposition** — catching it in a real model where the features are unknown is listed as an open problem. “How many features are in superposition” has only loose compressed-sensing bounds. The interference-weight definitions are both intractable at scale and the heuristics are admittedly proxies. And the Adam attribution is provisional — “we cannot conclusively put the blame on Adam.”

The blind spot the corpus barely names. Every notion of “near-orthogonal,” “interference,” “angle,” and “cosine similarity” here is computed in the *standard* Euclidean inner product on a space the same authors insist has no privileged basis. The privileged-bases paper shows training breaks the gauge, but the broken metric is never *derived* — it is observed to exist and then used. So the entire interference story rests on a metric the architecture declares arbitrary and the optimizer happens to fix; the closed-form polytopes are polytopes *in that metric*.

Connections

To Group 3 (SAEs as the inverse problem). Group 2 asks how a model *packs* n sparse features into m dimensions (a forward statement about W , upper-bounded by compressed sensing); Group 3 (Bricken et al. [3], Templeton et al. [8]) is the *recovery* algorithm compressed sensing promises — learn an overcomplete dictionary D and sparse code f with $x \approx Df$. Same theorem read in both directions (*Landscape* survey Part III.5). Toy Models even states the inverse problem explicitly as the dictionary-learning route to “solving superposition,” and Anthropic Interpretability Team [10] ties directly back, reproducing the logit-weight phenomenology of Bricken et al. [3].

To Group 4 (interference weights). The interference-weights paper *is* the bridge: its $U = W^*W$ virtual weights are exactly the **virtual weights** of Elhage et al. [1] (*ML Concepts* primer §1.3), and its real-vs-interference split is the obstruction Ameisen, Lindsey, et al. [9] designed attribution graphs to sidestep (per-prompt graphs dodge the global interference problem). The hope of one day doing *global* circuit analysis hinges on filtering interference weights — the open question the paper closes on.

To the gauge/metric theme. The cleanest statement the group offers a mathematician: **the existence of superposition is gauge-invariant; the cost is gauge-dependent.** Existence is $\ker W \neq 0$, untouched by $M \in GL(V)$. Interference is the off-diagonal of $G = W^*W$, which transforms under the metric and so is meaningful only after a choice of inner product — the very choice the residual stream withholds. The constructive, gauge-free alternative the *Landscape* survey

proposes (Part III.2) is to measure interaction not by cosine but by whether one component’s *image* lands in another’s *kernel* — the vanishing of the virtual weight $W_{\text{in}}^B W_{\text{out}}^A$. The interference-weights paper, working entirely in the metric-laden $U = W^*W$, is precisely the place that re-expression could be tried.

For the mathematician

This group is the richest deposit of classical mathematics in the corpus, and the place where a mathematician can both verify and extend.

Sphere-packing and the Thomson problem. The interference term of the 1-sparse loss is a generalized Thomson energy on S^{m-1} , and the empirically learned configurations are its low-order solutions — simplices, antipodal pairs, triangles, tetrahedra, the square antiprism. The *tegum-product* decomposition (orthogonal direct sums of uniform polytopes, with provably zero cross-factor interference) is a clean structural claim and an invitation: which higher-dimensional Thomson solutions factor as tegum products, and does the “stickiness” at rational dimensions-per-feature have an exact account? The polytope $\leftrightarrow$ low-rank-PSD correspondence ($G = W^*W$ as a Gram matrix of points) is the bijection that makes all of this rigorous.

Frames, polytopes, and uniqueness. Superposition is overcomplete-frame design: the columns of W are a unit-norm frame whose coherence one wants small. The “Mercedes-Benz frame” is the triangle solution; the regular-simplex solution $W \perp (1, \dots, 1)$ is the equiangular tight frame. The double-descent paper’s T -gons are cyclic frames. There is room to connect the learned optima to known optimal-frame and equiangular-tight-frame theory, and to ask when the optimum is unique versus a continuum.

Compressed-sensing phase transitions. The capacity bound $m = \Omega(-n(1 - S) \log(1 - S))$ is a faithful specialization of sparse-recovery theory, and the paper flags a genuine open analogy: the *first-order* phase transition in the toy model’s feature/neuron geometry versus the well-known recovery phase transition of compressed sensing (the Donoho–Tanner phase boundary). These are transitions in different quantities — per-feature representability versus whole-vector recoverability — and relating them precisely is unclaimed.

The under-examined metric. The deepest gap is conceptual and squarely in a geometer’s wheelhouse. Interference, coherence, polytope angles, kurtosis, and the interference weights $U - U^*$ are all computed in the standard inner product, while the residual stream carries a $GL(V)$ gauge with no canonical metric. Elhage et al. [4] shows training breaks the gauge but never derives the resulting metric; the rest of the group uses it implicitly. The open problems: derive the data-induced inner product the way the LRH paper derives $\text{Cov}(\gamma)^{-1}$ for the output space (Park et al. [7]); rewrite the Thomson/polytope geometry in that metric and ask which configurations survive; and re-express interference and interference weights in gauge-invariant (image-in-kernel) terms rather than through the $G = I$ Gram matrix. The whole beautiful geometry of this group is, at present, geometry in a metric nobody has justified.

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